

PlasticNet AI

The Intelligence Layer for Plastic Credit Markets & Circular Plastics

Executive Summary

The world produces 400 million tonnes of plastic annually, with less than 9% recycled. As regulators worldwide mandate Extended Producer Responsibility (EPR), plastic credits emerge as the carbon credits of the 2020s. **PlasticNet AI** is the definitive intelligence platform for plastic credit markets, waste stream optimization, and verified circular plastics — enabling brands, recyclers, and governments to trade, track, and transform the \$600 billion plastic waste crisis into a functioning market.

The Opportunity: Plastic credits are projected to become a \$100B+ market by 2030. But unlike carbon, there's no trusted infrastructure for verification, trading, or intelligence. PlasticNet AI becomes the Bloomberg Terminal + Verra + Stripe for the plastic economy.

The Problem

1. The Plastic Pollution Crisis

- **400M tonnes** produced annually; **350M tonnes** become waste
- Only **9%** is recycled globally; 22% is mismanaged (landfills, oceans)
- **8 million tonnes** enter oceans annually
- Microplastics now found in human blood, breast milk, and brains

2. Regulatory Tsunami

- **EU Plastic Tax:** €0.80/kg on non-recycled plastic packaging (2021+)
- **EPR Laws:** 30+ countries mandate producers fund end-of-life costs
- **US State Laws:** California, Maine, Oregon require recycled content mandates
- **Corporate Commitments:** 500+ companies pledged 25% recycled content by 2025

3. The Verification Void

- No standardized plastic credit methodology (fragmented standards)
- Rampant greenwashing — brands buy dubious “plastic neutrality”
- Recyclers can't prove chain of custody
- EPR compliance is manual, expensive, error-prone

4. Market Inefficiency

- No transparent pricing for plastic credits
 - Recyclers underpaid; brands overpay intermediaries
 - Quality mismatches (food-grade vs. downcycled)
 - Geographic arbitrage invisible
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The Solution: PlasticNet AI

Core Platform Architecture

PlasticNet AI Platform

Verification Layer	Trading Infrastructure	Intelligence Engine
• Mass Balance • Chain of Custody • Material ID • Satellite/IoT	• Credit Issuance • Spot Trading • Forward Contracts • Settlement	• Price Discovery • Supply Forecasting • Regulatory Mapping • Portfolio Optimization
Sensors & APIs	Registry & Ledger	AI/ML Models

Product Suite

1. PlasticVerify — Verification-as-a-Service

- AI-powered mass balance tracking across supply chains
- Spectroscopic material identification (NIR/FTIR integration)
- Satellite imagery for informal sector waste tracking
- Blockchain-anchored chain of custody
- Fraud detection using anomaly models

2. PlasticExchange — The Trading Platform

- First regulated spot market for plastic credits
- Multiple credit types: Collection, Recycling, Recycled Content
- Geographic and polymer-specific pricing
- Forward contracts for long-term offtake
- API-first for automated compliance purchasing

3. PlasticIQ — Market Intelligence

- Real-time pricing indices by region, polymer, grade
- Regulatory tracker (EPR deadlines, tax rates, mandates)
- Supply/demand forecasting
- Competitor benchmarking
- M&A and investment intelligence

4. PlasticComply — EPR Automation

- Automated reporting for 30+ jurisdictions
- Fee calculation and payment orchestration
- Audit-ready documentation
- Gap analysis and credit procurement

Market Opportunity

Total Addressable Market (TAM): \$180B

- **Plastic Credit Trading:** \$100B by 2030 (projected)
- **EPR Compliance Services:** \$25B annually
- **Recycling Intelligence:** \$15B

- **Circular Plastics Infrastructure:** \$40B

Serviceable Addressable Market (SAM): \$45B

- Global brands with plastic commitments
- Recyclers and waste management companies
- Government EPR schemes
- Impact investors and credit traders

Serviceable Obtainable Market (SOM): \$2.5B by 2030

- Platform fees: 2-5% of traded volume
- SaaS subscriptions: \$50K-500K/year
- Data and intelligence: \$25K-200K/year

Business Model

Revenue Streams

Stream	Model	Pricing
PlasticExchange	Transaction Fee	2-5% per trade
PlasticVerify	Usage-based	\$0.10-\$5 per verification
PlasticIQ	SaaS	\$25K-500K/year
PlasticComply	SaaS + Managed Service	\$50K-1M/year
Credit Issuance	Origination Fee	10-15% of credit value

Unit Economics

Metric	Target
Gross Margin	75-85%
Net Revenue Retention	130%+
CAC Payback	12 months
LTV/CAC	5x+

Competitive Landscape

Plastic Credit Standards (Not Platforms)

- **Verra Plastic Program** — Methodology only, no trading
- **Plastic Credit Exchange** — Early, limited volume
- **Second Nature** — Consumer-focused, small scale

Waste Management Tech

- **Rubicon** — Hauler marketplace, not credits
- **AMP Robotics** — Sorting AI, not trading
- **Greyparrot** — Waste analytics, not end-to-end

Gap

No one provides: Verification + Trading + Intelligence + Compliance in one platform.

PlasticNet AI is the **first vertically integrated plastic market infrastructure**.

Technology Differentiation

AI/ML Capabilities

1. Material Fingerprinting

- Computer vision + spectroscopy for polymer identification
- Grade assessment (food-grade, downcycle, etc.)
- Contamination detection

2. Mass Balance AI

- Real-time tracking across complex supply chains
- Probabilistic allocation models
- Audit trail generation

3. Satellite + IoT Integration

- Landfill and ocean leakage monitoring
- Informal sector collection verification
- Real-time waste flow mapping

4. Price Prediction

- Market-making algorithms
- Demand forecasting from regulatory calendars
- Arbitrage opportunity detection

5. Fraud Detection

- Anomaly detection for double-counting
 - Network analysis for circular trades
 - Document verification
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Go-to-Market Strategy

Phase 1: Foundation (Months 1-12)

Target: 50 enterprise customers, \$5M ARR

- Launch with 3 anchor CPG brands (P&G, Unilever, Nestlé-type)
- Partner with 2 major recyclers per region
- Integrate with existing MRFs (Material Recovery Facilities)
- Pilot in EU (strongest regulatory pressure)

Phase 2: Liquidity (Months 12-24)

Target: 200 customers, \$25M ARR, \$500M trading volume

- Launch PlasticExchange with market makers

- Expand to US (CA, OR, ME EPR markets)
- Add forward contracts and derivatives
- Launch PlasticIQ data products

Phase 3: Scale (Months 24-36)

Target: 500 customers, \$75M ARR, \$5B trading volume

- Global expansion (APAC, LATAM)
 - Government EPR scheme integrations
 - Launch PlasticComply managed services
 - Strategic acquisition of regional registries
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Traction & Validation

Market Signals

- **Plastic credits traded:** \$50M (2023) → \$500M (2025 est.)
- **Corporate demand:** 500+ companies committed to recycled content
- **Policy momentum:** EU, UK, US, India, Australia mandating EPR

Early Partnerships (Target)

- **Brands:** CPG companies with plastic commitments
 - **Recyclers:** Veolia, Waste Management, Republic Services
 - **Standards:** Verra, Ellen MacArthur Foundation
 - **Governments:** EU Plastics Tax administrators
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Team Requirements

Founding Team

- **CEO:** Commodity markets + climate tech background
- **CTO:** Supply chain + blockchain + AI expertise
- **CPO:** Environmental markets or carbon credit platforms
- **CCO:** Enterprise sales to CPG/retail

Key Hires (Year 1)

- Head of Standards (ex-Verra, Gold Standard)
 - Head of Recycler Partnerships
 - ML Lead (supply chain optimization)
 - Regulatory Affairs (EU, US, APAC)
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Funding Strategy

Seed Round: \$8M

- MVP development
- Initial pilot customers
- Team of 15
- 18-month runway

Series A: \$30M (Month 18)

- Platform launch
- EU + US expansion
- Team of 50
- Marketing and partnerships

Series B: \$80M (Month 36)

- Global scale
- Acquisitions
- Team of 150
- Path to profitability

Financial Projections

Year	ARR	Trading Volume	Customers	Team
1	\$5M	\$100M	50	25
2	\$25M	\$1B	200	60
3	\$75M	\$5B	500	120
4	\$150M	\$15B	1,000	200
5	\$300M	\$40B	2,000	300

Risk Analysis

Risks & Mitigations

Risk	Probability	Impact	Mitigation
Regulatory fragmentation	High	Medium	Multi-standard support, policy engagement
Greenwashing backlash	Medium	High	Rigorous verification, transparency
Market liquidity	High	High	Market maker partnerships, volume incentives
Incumbent competition	Medium	Medium	First-mover, vertical integration
Technology complexity	Medium	Medium	Modular architecture, partnerships

Why Now?

1. Regulatory Forcing Function

- EU Plastic Tax live; EPR mandates proliferating
- US state laws creating compliance urgency
- 2025-2030 corporate commitment deadlines

2. Market Creation Moment

- Plastic credits at inflection point (like carbon 2005-2010)
- First-mover can set standards and capture liquidity

3. Technology Readiness

- AI/ML mature for supply chain tracking
- IoT sensors affordable for waste streams
- Blockchain suitable for credit registries

4. Capital Availability

- Climate tech funding at all-time highs
- Plastic specifically gaining investor attention
- Infrastructure plays attracting growth capital

The Vision

By 2030, PlasticNet AI aims to:

- **Process \$50B+ in annual plastic credit trading volume**
- **Track 50% of global plastic waste flows**
- **Power EPR compliance for 100+ jurisdictions**
- **Enable 500+ brands to achieve verified circular plastics**

We're not just building a platform — we're building the **financial infrastructure that makes the circular plastics economy possible.**

Investment Thesis

PlasticNet AI is the Bloomberg + Verra + Stripe of the emerging \$100B plastic credit market.

The plastic pollution crisis is becoming the climate crisis of the 2020s. As regulation forces action and markets emerge, the company that provides trusted verification, liquid trading, and compliance automation will capture extraordinary value.

We're that company.

“The plastic crisis won't be solved by recycling alone. It requires markets. PlasticNet AI builds the infrastructure for those markets to function.”

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